```
...
}
```

The example is almost identical to the previous one. Except for the ampersand characters.

```
$ php swap2.php
Outside the swap function:
$a is 4
$b is 7
Inside the swap function:
$a is 7
$b is 4
Outside the swap function:
$a is 7
$b is 4
```

Here we see that the swap() function really changed the values of the variables.

3.1.9 PHP function recursion

Keyword

Factorial of a non-negative integer n, denoted by n!, is the product of all positive integers less than or equal to n.

Recursion, in mathematics and computer science, is a method of defining functions in which the function being defined is applied within its own definition. In other words, a recursive function calls itself to do its job. Recursion is an alternative to iteration. Recursion is the main approach in functional languages.

A typical example is the calculation of the **factorial**.

recursion.php

```
<?php

function factorial($n) {

    if ($n==0) {

        return 1;
    } else {
```